

VAPO: Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024

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Base Model: Qwen2.5-32B

Why Value Models? Finer Credit Assignment

Value-Model-Free (GRPO, DAPO)

- **Uniform Credit Assignment:** Assigns advantage based on final trajectory reward uniformly across all tokens.
- **Limitation for CoT:** Cannot trace each action's impact on returns.
- **Vulnerability:** Subtle step-level errors can cause catastrophic failures without precise feedback.

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Value-Model-Based (VAPO)

- **Precise Token-Level Credit:** Enables tracing each action's impact on final returns directly.
- **Variance Reduction:** Provides lower-variance estimates per token compared to Monte Carlo methods.
- **Better Generalization:** A well-trained value model generalizes across samples, improving sample efficiency.

Three Challenges in Value-Model-Based RL

Plaguing Value-Model-Based RL in Long-CoT Tasks

1

Value Model Bias

Stems from long trajectories and bootstrapped learning. Early errors compound over long reasoning chains.

2

Heterogeneous Lengths

Responses have highly variable lengths, requiring different bias-variance tradeoffs that standard methods fail to balance.

3

Reward Sparsity

Only the final <eos> token receives a reward signal, exacerbated by long Chain-of-Thought (CoT) processes.

VAPO Framework: Integrated Solutions

Component	Addresses Challenge
Value-Pretraining	Pre-trains value model to reduce initial bias.
Decoupled-GAE	Separates GAE computation to handle value estimation robustly
Length-adaptive GAE	Dynamically adjusts λ based on length for bias-variance tradeoff
Clip-Higher	Asymmetric clipping encouraging exploration.
Token-level Loss	Normalizes loss at the token level, not sequence level.
Positive Example LM Loss	Self-imitation learning on correct trajectories.
Group-Sampling	Samples multiple responses for better baseline estimation.

Length-Adaptive GAE

Dynamic λ Adjustment for Variable Response Lengths

Short Responses

Lower λ

Lower Variance

Relies more on the value model for stable learning.



Long Responses

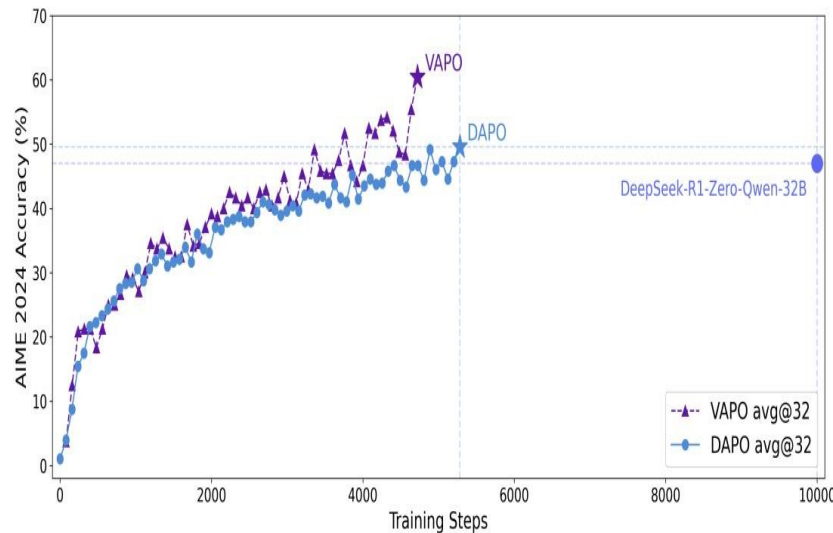
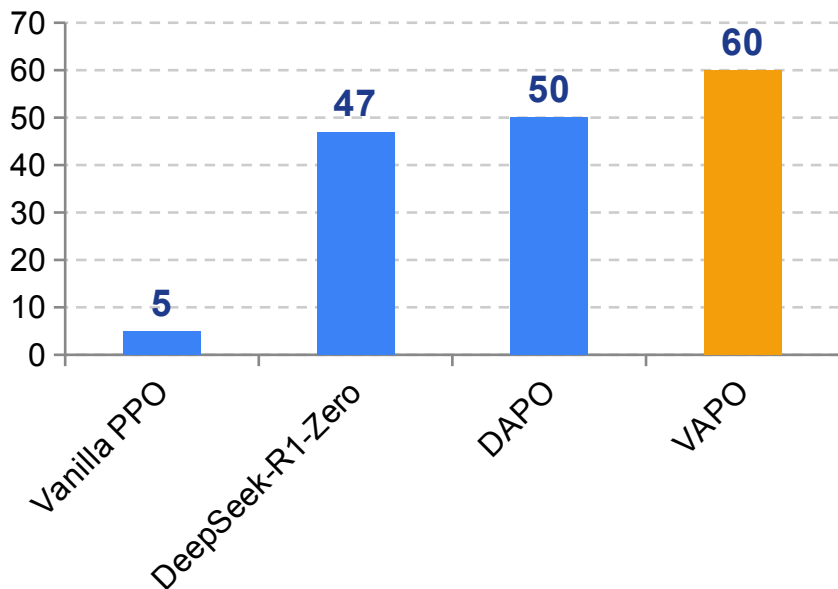
Higher λ

Lower Bias

Trusts empirical returns more to avoid compounding errors from the value model.

VAPO Scores 60 on AIME 2024

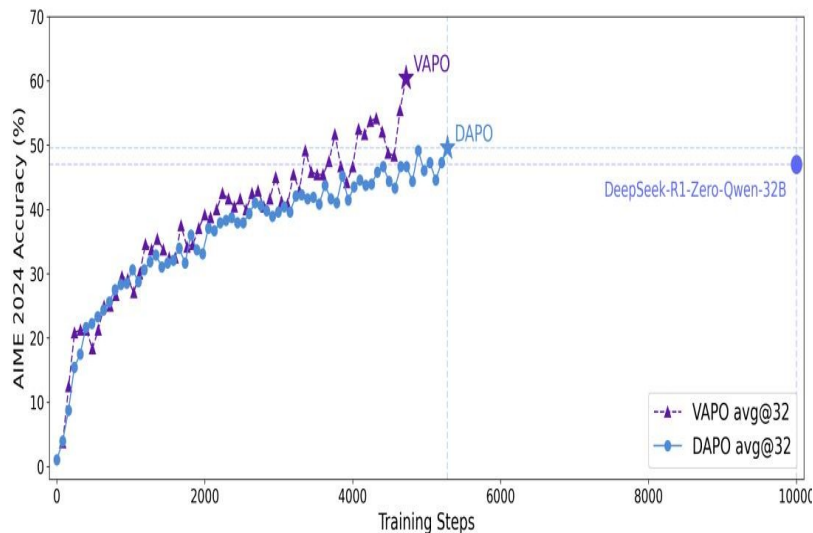
10+ Points Above DAPO and DeepSeek-R1-Zero



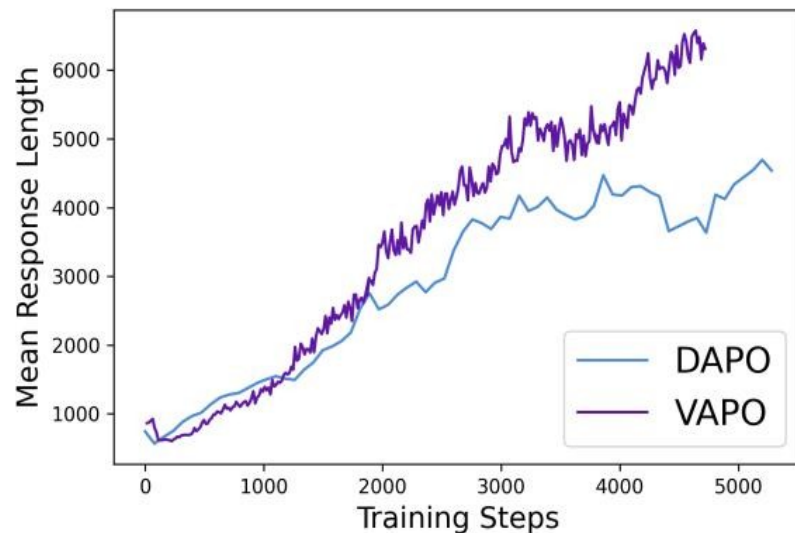
Training curve shows VAPO consistently outperforming DAPO throughout training.

VAPO Reaches SOTA in 5K Steps

Half the Training Budget of DAPO with Zero Training Crashes



*VAPO reaches 60% accuracy around step 4,500.
DAPO plateaus near 50% at 5,500 steps.*



VAPO generates longer reasoning chains (~6,000 tokens vs ~4,500 for DAPO).

Ablation: Every Component Matters

Value-Pretraining Alone Adds 49 Points (11 to 60)

Component Removed	Score (AIME 2024)	Impact
None (Full VAPO)	60	-
Group-Sampling	55	-5 pts
Positive Example LM Loss	54	-6 pts
Token-level Loss	53	-7 pts
Clip-Higher	46	-14 pts
Length-adaptive GAE	45	-15 pts
Decoupled-GAE	33	-27 pts
Value-Pretraining	11	-49 pts

Key Takeaways

Value-Model-Based RL Is Back — With the Right Engineering

- 1. Superior Performance on Reasoning:** Value-model-based methods surpass value-free when bias, heterogeneous lengths, and reward sparsity are solved.
- 2. Length-Adaptive GAE is Critical:** Dynamic bias-variance tradeoffs are required for heterogeneous reasoning chains.
- 3. Pretraining is Non-Negotiable:** Value pretraining is the single most important component. Without it, value-model-based RL barely functions.
- 4. Highly Stable and Efficient:** Reaches SOTA in ~5,000 steps with zero training crashes across multiple runs.
- 5. Future of LLM Reasoning:** Demonstrates that precise, token-level credit assignment is the optimal path forward for complex CoT tasks.